Jan 2020

Q3 Matrices & Recursion

a) Identity Matrix I

$2\times2 \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$3\times3 \quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

b) $A = \begin{bmatrix} 4 & -2 \\ 2 & -2 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 3 \\ 1 & 2 \end{bmatrix} \quad C = \begin{bmatrix} 3 & -1 \end{bmatrix}$

i) Calculate $2C(B-A)$

$$2C\left(\begin{bmatrix} -1 & 3 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 4 & -2 \\ 2 & -2 \end{bmatrix}\right)$$

$$= 2C \begin{pmatrix} -5 & 5 \\ -1 & 0 \end{pmatrix}$$

$$= 2(3 \ -1)\begin{pmatrix} -5 & 5 \\ -1 & 0 \end{pmatrix}$$

$$= \begin{bmatrix} 6 & -2 \end{bmatrix}\begin{bmatrix} -5 & 5 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 6\times-5 + -2\times-1 & 6\times5 + -2\times0 \end{bmatrix}$$

$$= \begin{bmatrix} -28 & -22 \end{bmatrix}$$

$1\times2 \quad 2\times2$

ii) determinant and ~~inverse~~ inverse of A

$$\det A = ad - bc$$

$$A^{-1} = \frac{1}{\det A}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A = \begin{bmatrix} 4 & -2 \\ 2 & -2 \end{bmatrix} \qquad \det A = (4)(-2) - (-2)(2) = -4$$

$$A^{-1} = \frac{1}{-4}\begin{bmatrix} -2 & 2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -1 \end{bmatrix}$$

c) $$D = \begin{bmatrix} -1 & 3 & 6 \\ 1 & 2 & 0 \\ 0 & 3 & 2 \end{bmatrix}$$

$$\det D = d_{11}C_{11} + d_{12}C_{12} + d_{13}C_{13}$$

$$A^{-1} = \frac{1}{\det D} C^T$$

get all cofactors.

$$d_{11}: \begin{bmatrix} -1 & 3 & 6 \\ 1 & 2 & 0 \\ 0 & 3 & 2 \end{bmatrix} \quad \begin{vmatrix} 2 & 0 \\ 3 & 2 \end{vmatrix} = (2)(2) - (0)(3) = 4$$

$$M_{11} = 4$$

$$C_{11} = M_{11} = 4$$

$$d_{12}: \begin{bmatrix} 1 & 3 & 6 \\ 1 & 2 & 0 \\ 0 & 3 & 2 \end{bmatrix} \quad \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} = (1)(2) - (0)(0) = 2$$

$$M_{12} = 2$$

$$C_{12} = -M_{12} = -2$$

d_{13} : $\begin{bmatrix} -1 & 3 & 6 \\ 1 & 2 & 0 \\ 0 & 3 & 2 \end{bmatrix}$ $\begin{vmatrix} 1 & 2 \\ 0 & 3 \end{vmatrix} = (1)(3) - (2)(0)$
$= 3$

$M_{13} = 3$

$C_{13} = M_{13} = 3$

d_{21} : $\begin{bmatrix} -1 & 3 & 6 \\ 1 & 2 & 0 \\ 0 & 3 & 2 \end{bmatrix}$ $\begin{vmatrix} 3 & 6 \\ 3 & 2 \end{vmatrix} = (3)(2) - (6)(3)$
$= -12$

$M_{21} = -12$

$C_{21} = -M_{21} = -(-12) = 12$

d_{22} : $\begin{bmatrix} -1 & 3 & 6 \\ 1 & 2 & 0 \\ 0 & 3 & 2 \end{bmatrix}$ $\begin{vmatrix} -1 & 6 \\ 0 & 2 \end{vmatrix} = (-1)(2) - (6)(0)$
$= -2$

$M_{22} = -2$

$C_{22} = M_{22} = -2$

d_{23} : $\begin{bmatrix} -1 & 3 & 6 \\ 1 & 2 & 0 \\ 0 & 3 & 2 \end{bmatrix}$ $\begin{vmatrix} -1 & 3 \\ 0 & 3 \end{vmatrix} = (-1)(3) - (3)(0)$
$= -3$

$M_{23} = -3$

$C_{23} = -M_{23} = -(-3) = 3$

d_{31} : $\begin{bmatrix} -1 & 3 & 6 \\ 1 & 2 & 0 \\ 0 & 3 & 2 \end{bmatrix}$ $\begin{vmatrix} 3 & 6 \\ 2 & 0 \end{vmatrix} = (3)(0) - (6)(2)$
$= -12$

$M_{31} = -12$

$C_{11} = M_{31} = -12$

$$d_{32}: \begin{bmatrix} -1 & 3 & 6 \\ 1 & 2 & 0 \\ 0 & 3 & 2 \end{bmatrix} \quad \begin{vmatrix} -1 & 6 \\ 1 & 0 \end{vmatrix} = (-1)(0) - (6)(1) = -6$$

$$M_{32} = -6$$

$$C_{32} = -M_{32} = -(-6) = 6$$

$$d_{33}: \begin{bmatrix} -1 & 3 & 6 \\ 1 & 2 & 0 \\ 0 & 3 & 2 \end{bmatrix} \quad \begin{vmatrix} -1 & 3 \\ 1 & 2 \end{vmatrix} = (-1)(2) - (3)(1) = -5$$

$$M_{33} = -5$$

$$C_{33} = M_{33} = -5$$

$$\det D = d_{11}C_{11} + d_{12}C_{12} + d_{13}C_{13}$$

$$= (-1)(4) + (3)(-2) + (6)(3)$$

$$= -4 - 6 + 18$$

$$= 8$$

$$C = \begin{bmatrix} 4 & -2 & 3 \\ 12 & -2 & 3 \\ -12 & 6 & -5 \end{bmatrix} \qquad C^T = \begin{bmatrix} 4 & 12 & -12 \\ -2 & -2 & 6 \\ 3 & 3 & -5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{8}\begin{bmatrix} 4 & 12 & -12 \\ -2 & -2 & 6 \\ 3 & 3 & -5 \end{bmatrix} = \begin{bmatrix} \frac{4}{8} & \frac{12}{8} & \frac{-12}{8} \\ \frac{-2}{8} & \frac{-2}{8} & \frac{6}{8} \\ \frac{3}{8} & \frac{3}{8} & \frac{-5}{8} \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{3}{2} & \frac{-3}{2} \\ \frac{-1}{4} & \frac{-1}{4} & \frac{3}{4} \\ \frac{3}{8} & \frac{3}{8} & \frac{-5}{8} \end{bmatrix}$$

d) See Notes on recursion for definition.

e) 2, 4, 6, 8, 10, 12, ...

non-recursive : $t(n) = 2n$

i.e. when $n=1$ $t(1) = 2(1) = 2$
$n=2$ $t(2) = 2(2) = 4$
$n=3$ $t(3) = 2(3) = 6$
⋮

recursive definition : $t(1) = 2$
$t(n) = t(n-1) + 2 \quad (n>1)$

i.e. when $n=2$ $t(2) = t(2-1) + 2$
$= t(1) + 2$
$= 2 + 2 = 4$
⋮